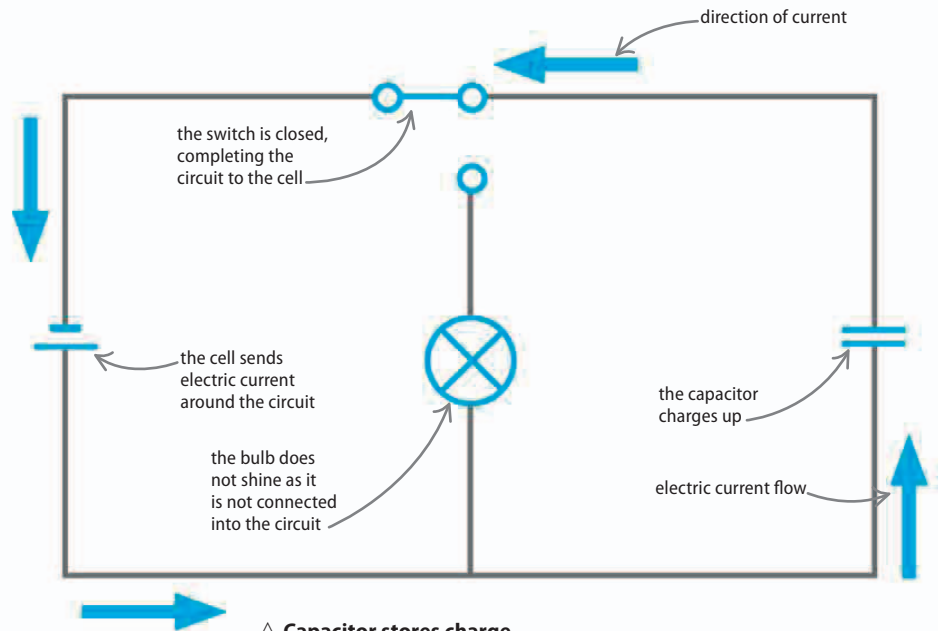


Capacitor

A capacitor is a component used in many circuits to store and release electric charge. There are many different types and sizes of capacitor, many of which are used in circuits to smooth out a varying electric current. At its simplest, a capacitor may consist of two plates of electrically conductive material separated by an insulator called a dielectric. In a direct current (DC) circuit (see page 216), the capacitor stops the flow of current once it has been fully charged.

Supercapacitors that store and release large electrical charges are being developed to replace electric vehicle batteries, since they can be **recharged far more quickly** and more often.



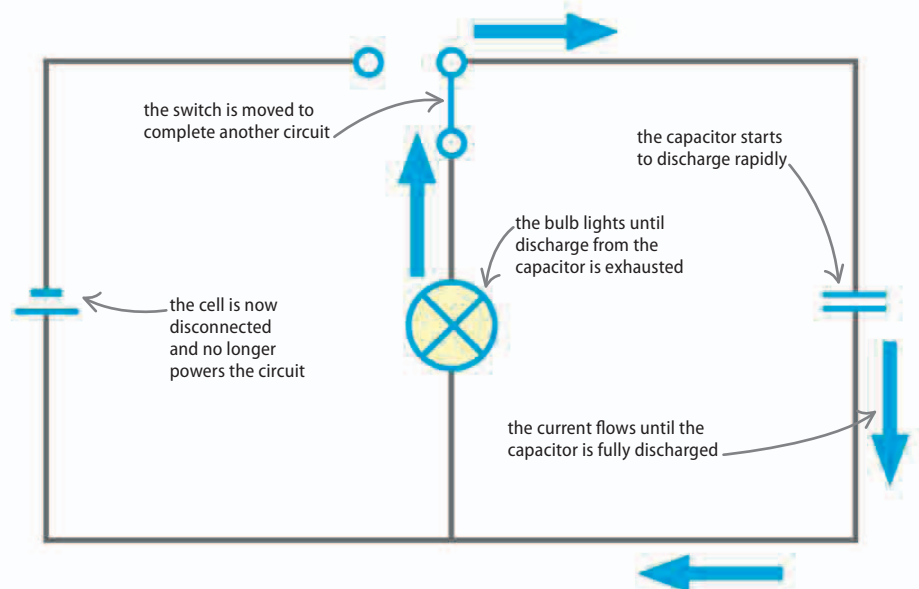
△ Capacitor stores charge

In this direct current (DC) circuit, charge flows from the cell to the capacitor. The electric charge builds up on the capacitor's plate while some current continues to flow across the capacitor and around the circuit. As its charge builds, the capacitor resists the flow of current. Once fully charged, it completely stops the flow of current through it.

REAL WORLD

Camera flash

Some capacitors are used because they can release their entire charge in just a fraction of a second. Most digital cameras use capacitors, which are charged up by the camera's battery, to power their flash function. The capacitor releases all of its charge almost instantly to enable the flash to fire brightly so that it lights up a dim scene as a photo is taken.



△ Capacitor releases charge

Moving the switch disconnects the cell from the circuit but closes and completes another circuit that still contains the capacitor. The capacitor discharges (releases its electrical charge) and the bulb lights up. The bulb will only shine for a short while and will stop once the capacitor is fully discharged.